

TOP CHALLENGES

- ▶ Most centres are not optimised for AI workloads
- ▶ Availability of low-cost power and uninterrupted power. Zero downtime needs additional power backups
- ▶ Lack of skilled workforce with expertise in emerging tech like AI GPUs and liquid cooling
- ▶ Regulatory challenges and clearances
- ▶ Global supply chain blockages for key components impacting deployments
- ▶ High capital expenditure and execution risks for large-scale projects which require significant upfront investment and long development periods

THE SUSTAINABILITY PUSH

Several data centre operators are working towards net-zero by 2030. Some initiatives include:



- ▶ CtrlS has set up a 137.5 MW solar farm in Nagpur and partnered with NTPC Green Energy to establish grid connected renewable energy projects. Its Mumbai facility uses grey water to cool its HVAC system



- ▶ Equinix has set up a captive solar plant and is building a wind farm



- ▶ Sify's 14 centres use around 38% renewable power



- ▶ NTT GDC has shifted to liquid and direct cooling methods at its Navi Mumbai data centre, resulting in a 30% improvement in energy efficiency at the plant



- ▶ Techno Digital sources 85% energy from renewable sources and has brought down water consumption by 75% through adiabatic cooling towers

ready infrastructure, which means designing campuses with higher power density and advanced cooling technologies.

COMBATING MANY CHALLENGES

The first change is the massive power requirements because of the compute capacity required. "The GPU infrastructure that we deploy for AI has a much higher rack density and draws a lot more power than traditional workloads," says Sharad Sanghi, Co-founder and CEO, Neysa, an AI-acceleration cloud provider. He has been closely involved with the evolution of this industry in India and points out that initially, data

centres would have a capacity of about 40 kW per rack depending on the number and the kind of servers. Now, this is going up to over 150 kW, so there is a significant shift in the energy density of the rack, which means conventional cooling techniques cannot be deployed. Neysa provides GPU compute, storage and network services, along with software and third-party integrations to enterprise customers, and currently uses NTT as a third-party data centre. Demand from neo-cloud providers is adding to the increased capacity requirements of the traditional operators. Neo-cloud companies are specialised, high-performance cloud providers

focusing on GPU-as-a-Service to power AI training and inference.

Hence, the basic build of a data centre needs to change, with higher floor clearance for the changing rack space. With power consumption going up significantly, the cooling methods are evolving as well.

Air cooling is effective only up to a certain power load; newer AI data centres will have to be liquid-cooled to remain efficient. This shift is opening an opportunity for newer players to make inroads into the segment.

Techno Digital, which recently launched its first facility in Chennai, is a spin-off from Techo Electric & Engineering Company,